Northrop Grumman Collaboration Project

Project Overview

Sponsored by:

NORTHROP GRUMMAN

By
Yen Nguyen

An Overview of
System Engineering

Presentation Overview

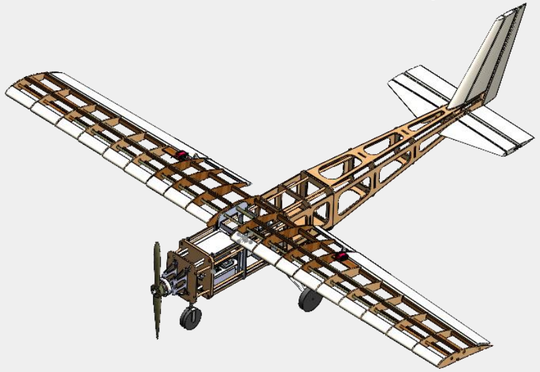
- Project Background
- Areas of Development
- Team Breakdown
- List of Abbreviation
- DoD Architecture Framework – Overview
- Mission Timeline
- Compliance Matrices
- Sequence Diagram
- Technical Performance Measures
- DoD Architecture Framework - Service Views
- Technical Risks

Project Background

- 15 years of Northrop Grumman support
- Started as a graduate student project
  - Freshman to graduate level
- Student-led with faculty and NG advisors
- Collaboration between Cal Poly Pomona (CPP) and San Luis Obispo (SLO)
- Project Goals:
  - Design and build autonomous systems in a rapid environment
  - Replicate industry standards

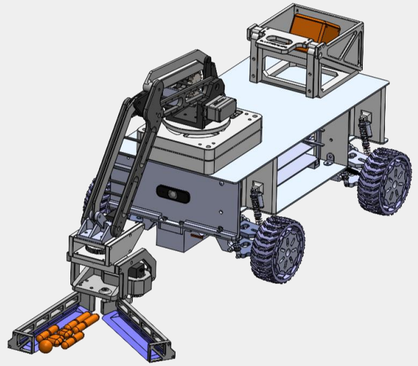
Primary Areas of Development

CPP Unmanned Aerial Vehicle
Multi-Role Aircraft (MRA)

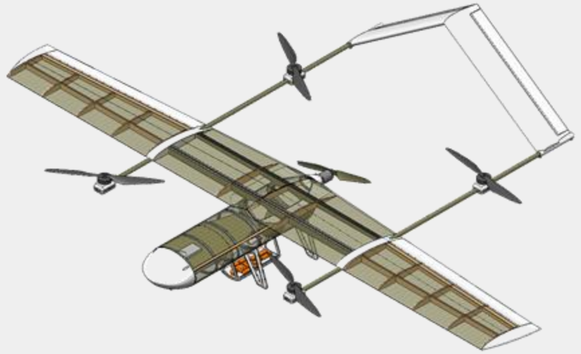


Ground Control Station (GCS)

CPP Unmanned Ground Vehicle
Emergency Response Unit (ERU)



CPSLO Unmanned Aerial Vehicle
Medical Evacuation Aircraft (MEA)



Work Breakdown Structure

Disaster Response System

Multi-Role Aircraft (MRA)

Structures
Manufacturing and Safety
Stability and Controls
Aerodynamics
Electrical
Payload Integration
Propulsion
Software

Emergency Response Unit (ERU)

Mechanical
- Chassis Drivetrain
- Payload
- Analysis

Electrical
- Digital Hardware Design
- Power
- Battery Management System

Software
- Firmware
- Autonomous
- Integration

Medical Evacuation Aircraft (MEA)

Aeronautics
Mechanical
Embedded Systems

Ground Control Station (GCS)

Infrastructure
Vehicle Integration
Database
User Interface

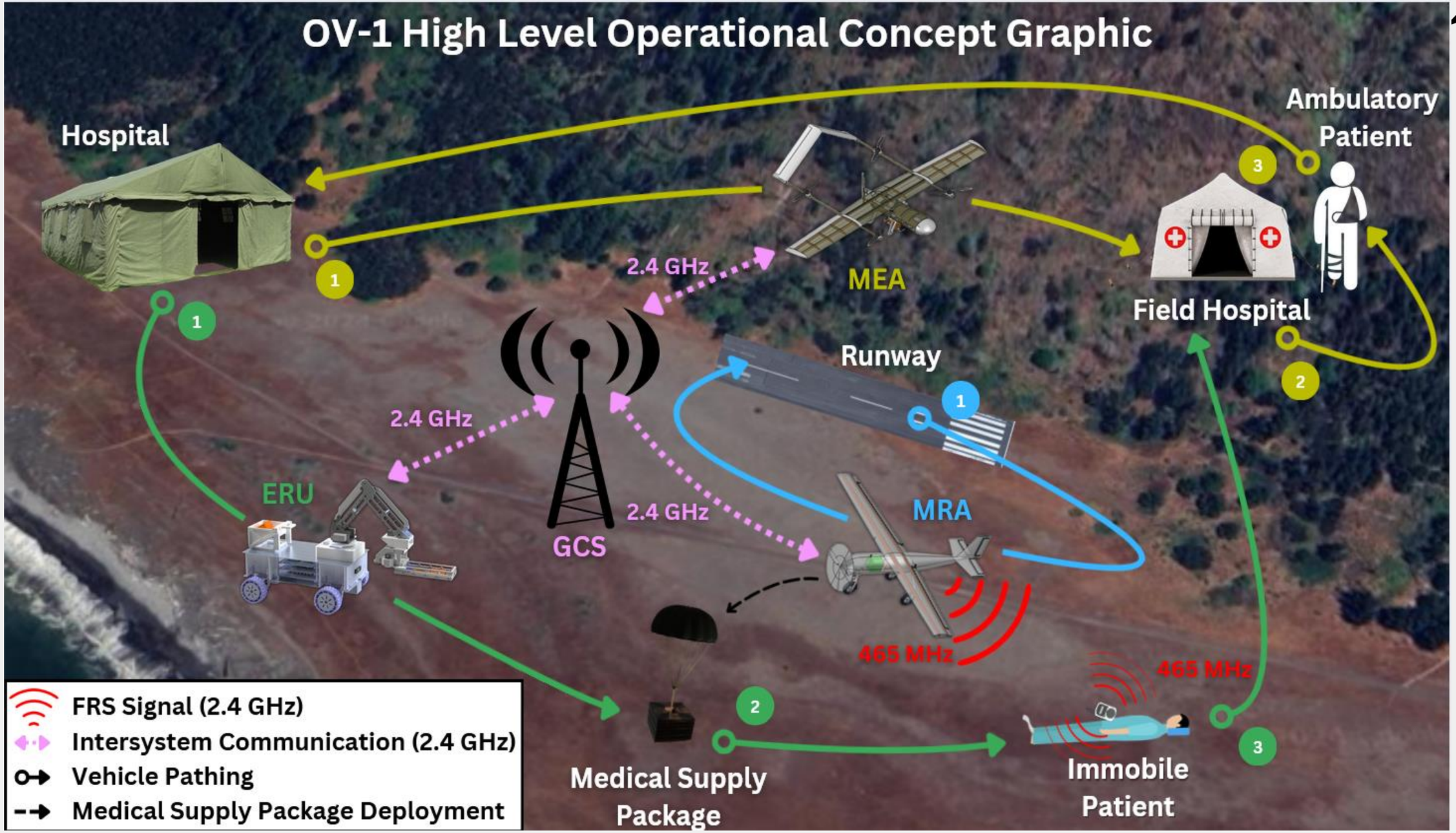
SE/Project Management

MRA
ERU
MEA
GCS
Events
Website Mgmt
Webmaster
Finance

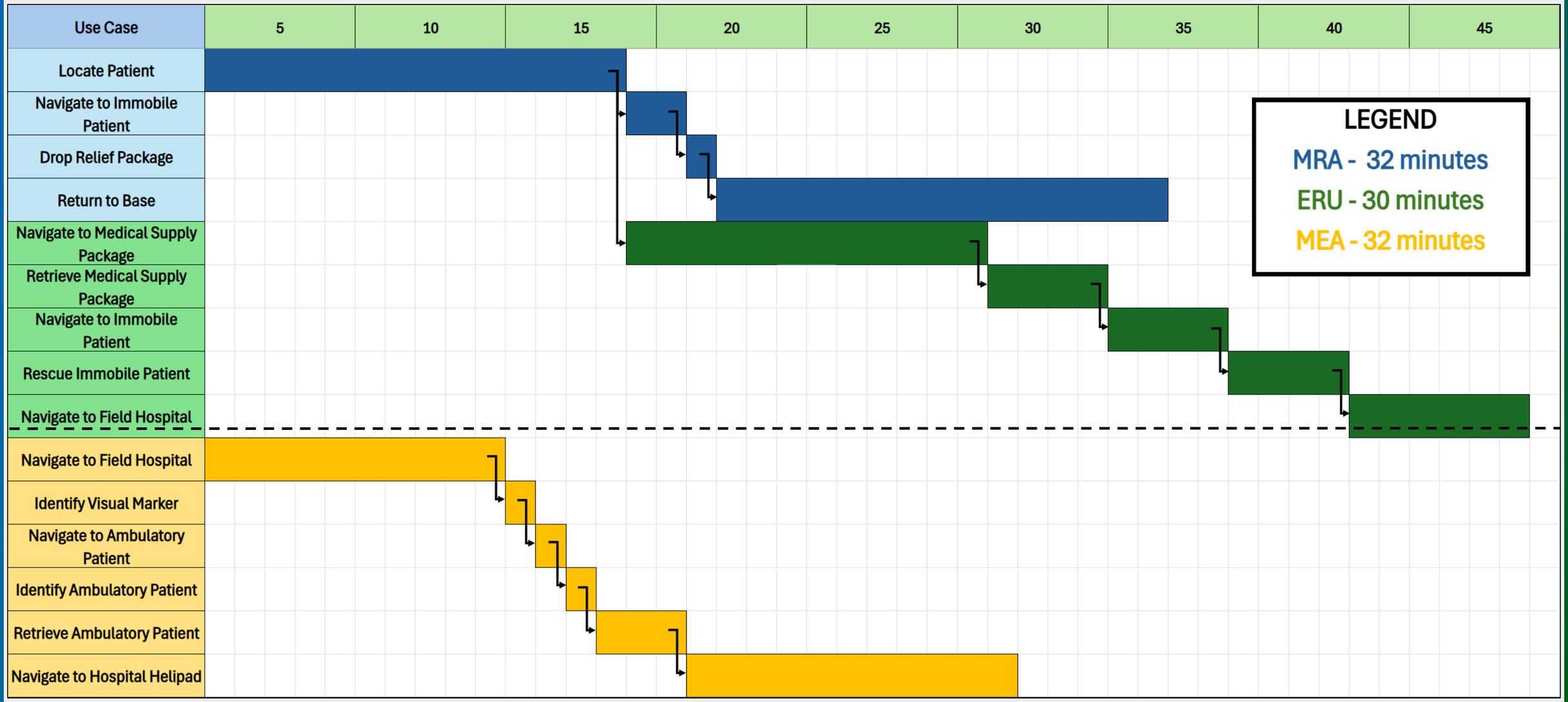
List of Abbreviations

- SE – Systems Engineering
- FRS – Family Radio Service
- TBD – To be Determined
- MRA – Multi-Role Aircraft
- ERU – Emergency Response Unit
- MEA – Medical Evacuation Aircraft
- GCS – Ground Control Station
- OV-1 – Operational Viewpoint - 1
- SV-1 – Systems Viewpoint - 1
- SV-2 – Systems Viewpoint - 2
- SV-4 – Systems Viewpoint - 4
- KPP – Key Performance Parameters
- TPM – Technical Performance Measures

OV-1 High Level Operational Concept Graphic



Overall Mission Timeline



Compliance Matrices

ID #	Requirement Text	Verification Method(s)	Verification Explanation	Status
[SYS_01]	The system SHALL be able to geolocate a walkie talkie broadcasting on Family Radio Service (FRS) channels to an accuracy of 10 ft.	Analysis, Search Test	Will perform an RF search using KrakenSDR and output the resulting data in a CSV file to measure accuracy compliance.	Expect to Comply
[SYS_02]	The system SHALL be capable of deploying a medical supply package.	Demonstration	On the ground, initiate payload deploy sequence to verify the payload can be dropped from the MRA.	Expect to Comply
[SYS_03]	The system SHALL be capable of transporting a medical supply package weighing 2.5lbs.	Flight Test, Drive Test	The system will be loaded with a 2.5lb payload and tested for successful transport.	Expect to Comply
[SYS_04]	The system SHALL be capable of transporting a medical supply package with dimensions 3in x 3in x 3in (L x W x H).	Demonstration	A 3in x 3in x 3in supply package will be placed in the MRA and ERU securement mechanisms to test the tolerance.	Expect to Comply

Compliance Matrices

ID #	Requirement Text	Verification Method(s)	Verification Explanation	Status
[SYS_05]	The medical supply package SHALL not experience an acceleration exceeding 2 G's during any phase of the mission.	Analysis, Drop Test	The release mechanism will be tested using an altimeter to verify the package experiences less than 2 G's.	Expect to Comply
[SYS_06]	The system SHALL transport an immobile patient from their location to predesignated field hospital on a medical stretcher.	Demonstration	In a mock run, the immobile patient will be manually secured by the ERU and autonomously navigated to waypoint to verify patient securement throughout transport.	Expect to Comply
[SYS_07]	The system SHALL be able to load a patient onto a medical stretcher provided by the system before any transportation begins.	Inspection, Demonstration	The ERU and MEA will be manually loaded with the patients and the securement will be verified.	Expect to Comply
[SYS_08]	The system SHALL be capable of transporting patients between 6 and 8 inches tall (inclusive).	Analysis, Flight Test, Drive Test	The stretcher mechanisms will be tested to ensure they can securely transport patients within the 6 to 8-inch height range.	Expect to Comply

Compliance Matrices

ID #	Requirement Text	Verification Method(s)	Verification Explanation	Status
[SYS_09]	The system SHALL be capable of transporting patients that weigh between 0.5 and 0.8 lbs (inclusive).	Analysis, Flight Test, Drive Test	The system will be tested by transporting simulated patients weighing between 0.5 and 0.8 lbs.	Expect to Comply
[SYS_10]	The system SHALL be capable of detecting a target area with dimensions no smaller then 3ft x 3ft (L x W).	Sensor Test	System's sensor placed in an operational environment and tasked with detecting and identifying a 3ft x 3ft target area.	Expect to Comply
[SYS_11]	The entire system SHALL be commanded and controlled from a single ground control station using one standardized communication network.	Analysis, Demonstration	In a mock run, GCS will control the ERU, MRA, and MEA using one communication network.	Expect to Comply

Compliance Matrices

ID #	Requirement Text	Verification Method(s)	Verification Explanation	Status
[SYS_12]	Once the immobile patient has been retrieved, the system shall verify that the patient is secure throughout the duration of travel.	Demonstration	In a mock run, once the immobile patient has been retrieved, the system shall demonstrate verification that the patient has been secured safely through the duration of travel.	Expect to Comply
[SYS_13]	Once the ambulatory patient has been retrieved, the system shall verify that the patient is secure throughout the duration of travel.	Demonstration	In a mock run once the ambulatory patient has been retrieved, the system shall demonstrate verification that the patient has been secured safely through the duration of travel.	Expect to Comply
[SYS_14]	The system SHALL be able to operate in weather conditions consisting of sunny to overcast, no rain to light rain (0.02 inch per hour), no wind to winds up to 5 mph	Analysis	The vehicles and GCS will be tested in simulated or natural weather conditions that span the permitted range.	Expect to Comply

Compliance Matrices

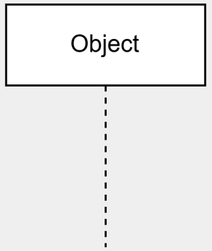
ID #	Requirement Text	Verification Method(s)	Verification Explanation	Status
[SYS_15]	All system components SHALL be able to be repaired and running within 10 minutes of component failure.	Demonstration	Vehicle teams will be timed as they demonstrate their rapid repair or replacement of the components most likely to fail (e.g. low battery).	Expect to Comply
[SYS_16]	The system SHALL be able to resist and recover from mission deviations and natural or external interference in real time without scrubbing or putting mission on hold.	Mock Run Test	In a mock run, the system shall measure the performance of all vehicles in adverse environment to verify resilience to mission deviations and external interference without mission interruptions.	Expect to Comply
[SYS_17]	The system SHALL be able to display and control all subsystems and monitor mission status through a Ground Control Station (GCS).	Inspection, Mock Run Test	The system shall test with mock runs the functioning of display and control of all subsystems and mission through GCS.	Expect to Comply

Compliance Matrices

ID #	Requirement Text	Verification Method(s)	Verification Explanation	Status
[SYS_18]	The system SHALL be able to complete all mission objectives within 60 minutes of mission start.	Mock Run Test	The system shall test through a mock run the execution of a timed mission scenario to ensure completion of all objectives within 60 minutes.	Expect to Comply
[SYS_19]	The system SHALL be capable of detecting visual markers with dimensions no smaller than 3ft x 3ft (L x W)	Flight Test	The MEA's computer vision will be tested on its ability to detect a 3ft x 3ft marker in operational conditions.	Expect to Comply
[SYS_20]	The system SHALL use standardized communication hardware across all system vehicles.	Inspection	All system vehicles will be inspected for standardized communication hardware.	Expect to Comply
[SYS_21]	The system SHALL be able to autonomously avoid obstacles to during all phases of patient transport.	Demonstration	Obstructions will be introduced in the System's autonomous navigation path to demonstrate obstacle avoidance.	Expect to Comply

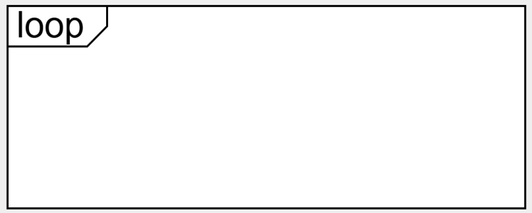
Sequence Diagram Legend

Lifeline

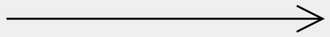


Activation

Loop Operator Fragment (events repeat)

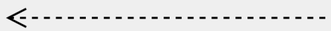


Asynchronous Message (no reply required)



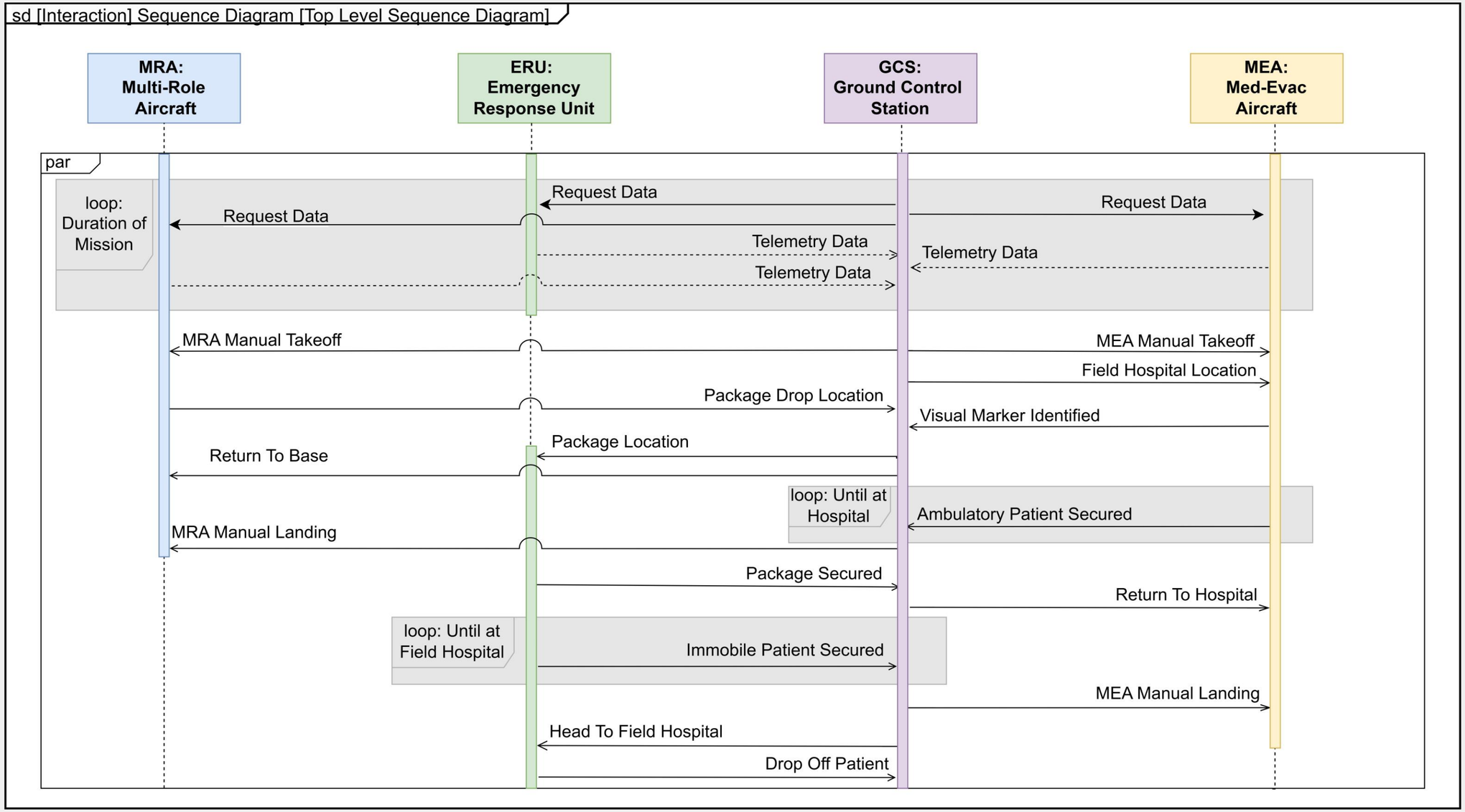
Synchronous Message (reply required)

Reply Message



Par Operator Fragment (events in parallel)

Sequence Diagram



TPMs & KPPs – Size

ID #	Team	TPM/KPP	Upper Bounds	Lower Bounds	Actual Values
[SYS_04]	ERU, MEA	The system SHALL be capable of transporting a medical supply package with dimensions 3in x 3in x 3in (L x W x H).	3 in x 3in x 3 in	3 in x 3 in x 3 in	3 in x 3 in x 3 in
[SYS_08]	ERU, MEA	The system SHALL be capable of transporting patients between 6 and 8 inches tall (inclusive).	8 in	6 in	TBD (ERU 4/25) (MEA 5/2)
[MRA_02]	MRA	The vehicle SHALL autonomously drop a medical supply package within a 250ft radius of the patient's geolocation.	250 ft	1 ft	TBD (4/18)
[ERU_02]	ERU	The vehicle SHALL be able to navigate water hazards of a minimum 3 in deep and 2 ft wide.	N/A	3 in deep, 2 ft wide	TBD (5/2)

TPMs & KPPs – Weight

ID #	Team	TPM/KPP	Upper Bounds	Lower Bounds	Actual Values
[SYS_03]	MRA, ERU	The system SHALL be capable of transporting a medical supply package weighing 2.5lbs.	2.5 lbs	2.5 lbs	2.5 lbs
[SYS_09]	ERU, MEA	The system SHALL be capable of transporting patients that weigh between 0.5 and 0.8 lbs (inclusive).	0.8 lbs	0.5 lbs	TBD (ERU 4/25) (MEA 5/2)
[MRA_10]	MRA	The vehicle weight SHALL not exceed 35 lbs unloaded.	35 lbs	N/A	TBD (4/18)
[ERU_15]	ERU	The vehicle WILL have a maximum weight of 35 lbs unloaded.	35 lbs	N/A	TBD (4/25)
[MEA_07]	MEA	The vehicle SHALL have a maximum take-off weight of 30 lbs.	30 lbs	N/A	TBD (5/2)

TPMs & KPPs – Performance

ID #	Team	TPM/KPP	Upper Bounds	Lower Bounds	Actual Values
[SYS_05]	SYS	The medical supply package SHALL not experience an acceleration exceeding 2 G's during any phase of the mission.	2 G's	N/A	1.77 G's
[ERU_01]	ERU	The vehicle SHALL be able to drive up a 30% grade hill at 1.5 mph while fully loaded.	N/A	1.5 mph	TBD (4/11)
[ERU_14]	ERU	The vehicle SHALL have a range of 1000ft fully loaded.	N/A	1000 ft	TBD (4/25)

TPMs & KPPs – Performance

ID #	Team	TPM/KPP	Upper Bounds	Lower Bounds	Actual Values
[SYS_01]	MRA	The system SHALL be able to geolocate a walkie talkie broadcasting on Family Radio Service (FRS) channels to an accuracy of 10 ft.	10 ft	N/A	TBD (4/18)
[GCS_18]	GCS	The GCS SHALL implement a physical communication layer capable of maintaining signal integrity on a field of dimensions no less than 1500ft x 1500ft (L x W)	N/A	1500ft x 1500ft	TBD (4/11)

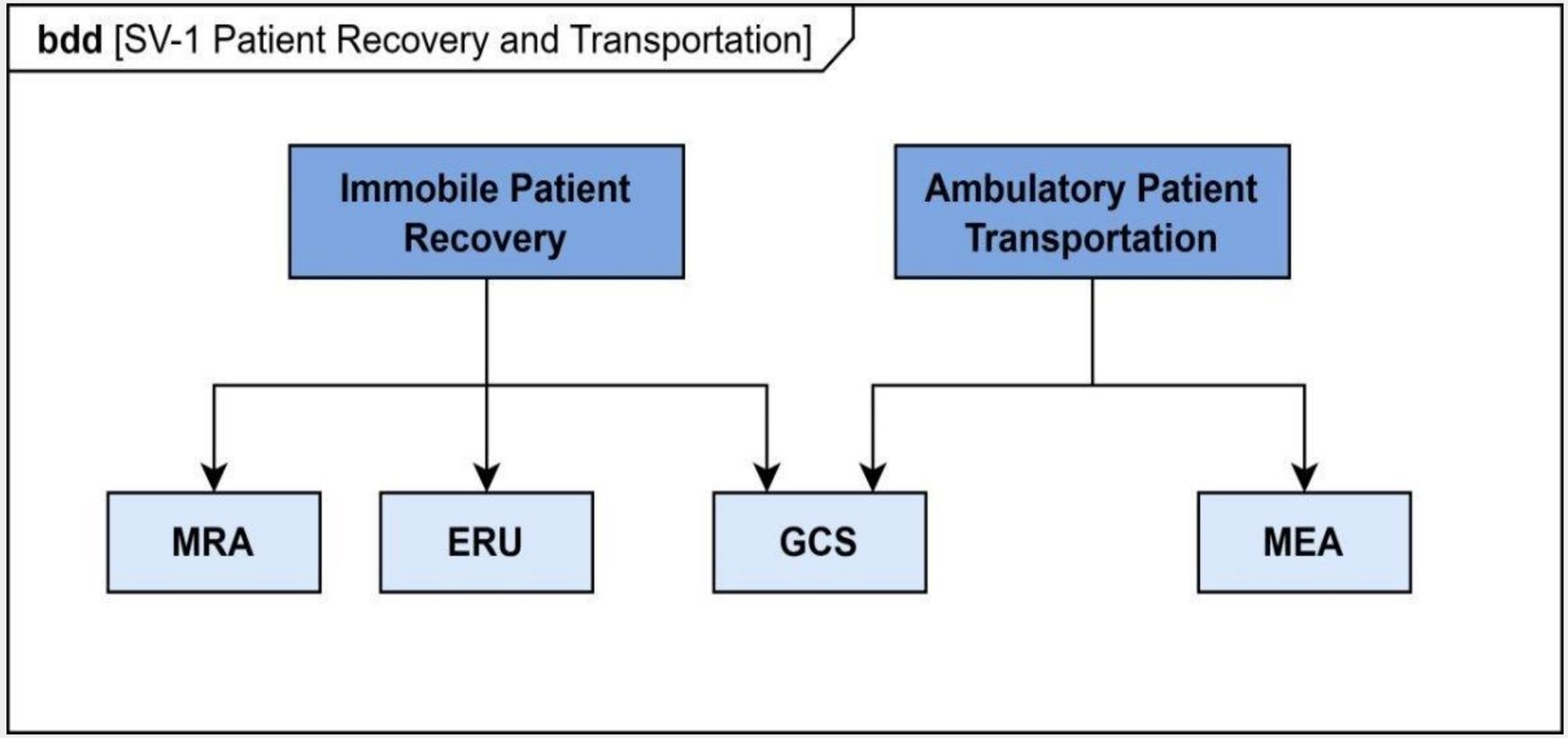
TPMs & KPPs – Computer Vision

ID #	Team	TPM/KPP	Upper Bounds	Lower Bounds	Actual Values
[SYS_10]	MEA	The system SHALL be capable of detecting a target area with dimensions no smaller then 3ft x 3ft (L x W).	N/A	3 ft x 3 ft	TBD (4/18)
[SYS_19]	MEA	The system SHALL be capable of detecting visual markers with dimensions no smaller than 3ft x 3ft (L x W)	N/A	3 ft x 3 ft	TBD (4/18)

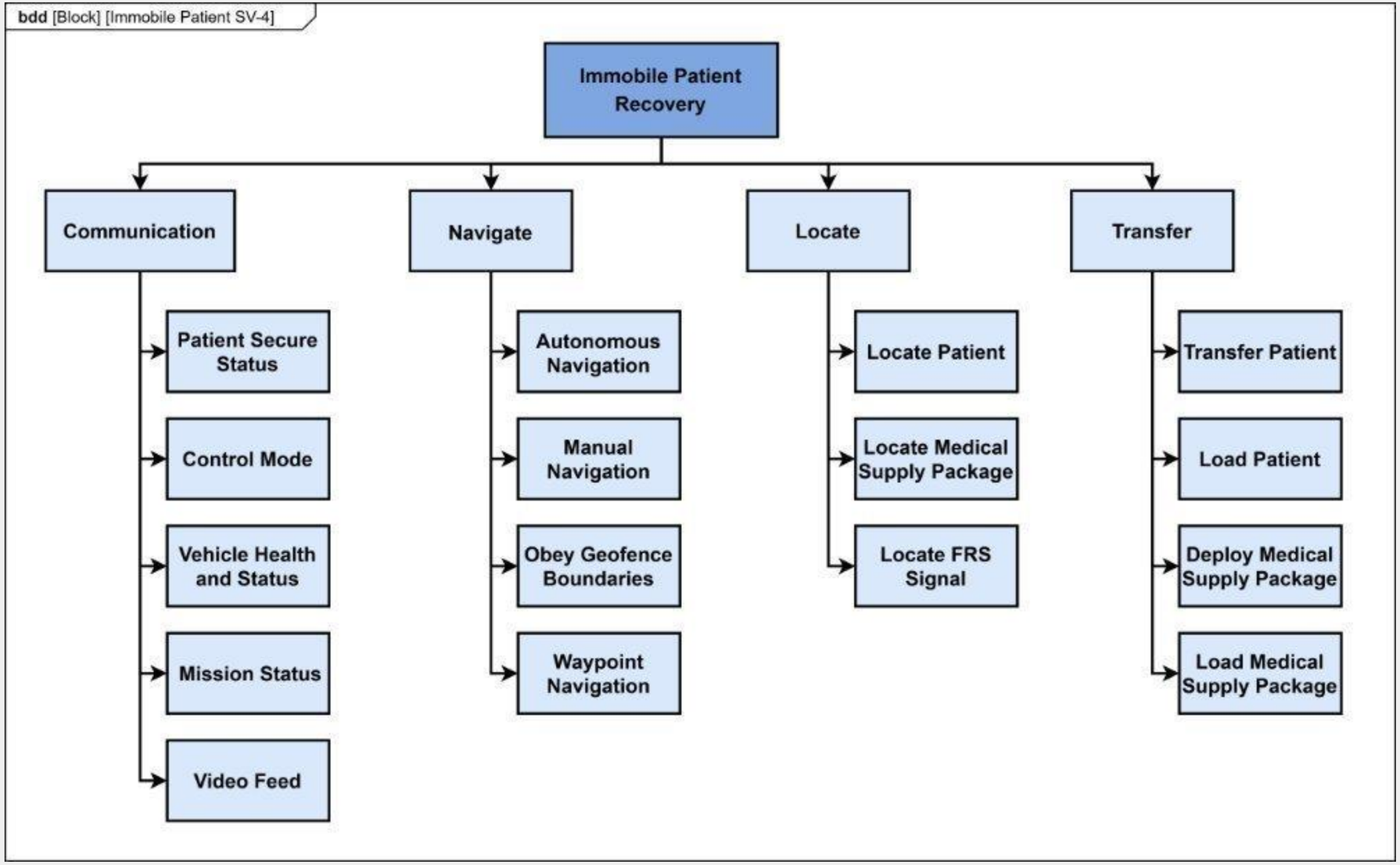
TPMs & KPPs – Flight Time

ID #	Team	TPM/KPP	Upper Bounds	Lower Bounds	Actual Values
[SYS_18]	MRA, MEA	The system SHALL be able to complete all mission objectives within 60 minutes of mission start.	60 mins	N/A	TBD (5/9)
[MRA_03]	MRA	The vehicle SHALL have a minimum, fully loaded, endurance flight time of 30 minutes.	N/A	30 mins	TBD (4/18)
[MEA_06]	MEA	The vehicle SHALL have a minimum, fully loaded, endurance flight time of 30 minutes.	N/A	30 mins	TBD (4/25)

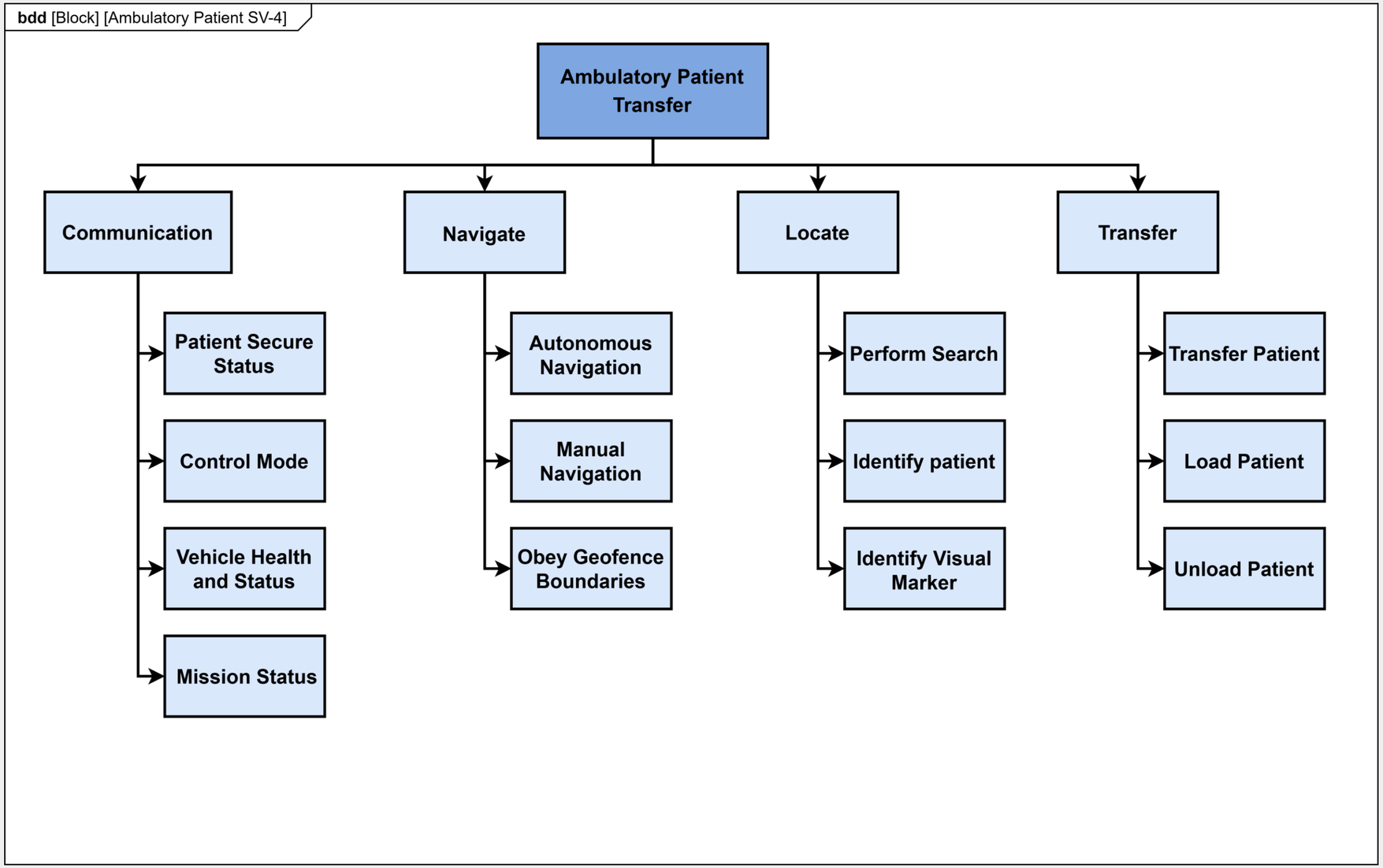
SV-1


bdd [SV-1 Patient Recovery and Transportation]
Immobile Patient Recovery
Ambulatory Patient Transportation
MRA
ERU
GCS
MEA

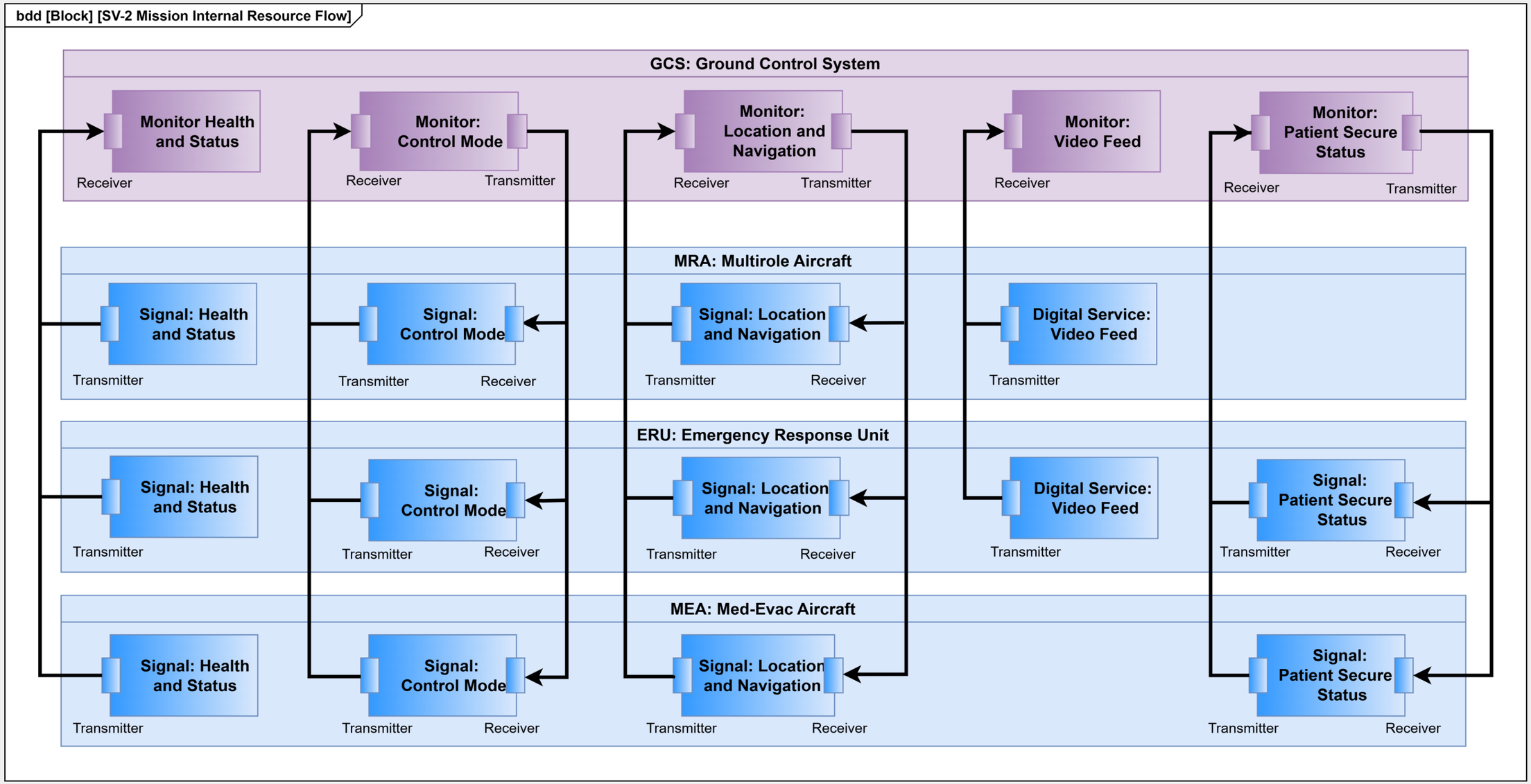
SV-4


bdd [Block] [Immobile Patient SV-4]
Immobile Patient Recovery
Communication
Navigate
Locate
Transfer
Patient Secure Status
Control Mode
Vehicle Health and Status
Mission Status
Video Feed
Autonomous Navigation
Manual Navigation
Obey Geofence Boundaries
Waypoint Navigation
Locate Patient
Locate Medical Supply Package
Locate FRS Signal
Transfer Patient
Load Patient
Deploy Medical Supply Package
Load Medical Supply Package

SV-4


bdd [Block] [Ambulatory Patient SV-4]
Ambulatory Patient Transfer
Communication
Navigate
Locate
Transfer
Patient Secure Status
Control Mode
Vehicle Health and Status
Mission Status
Autonomous Navigation
Manual Navigation
Obey Geofence Boundaries
Perform Search
Identify patient
Identify Visual Marker
Transfer Patient
Load Patient
Unload Patient

SV-2


bdd [Block] [SV-2 Mission Internal Resource Flow]
GCS: Ground Control System
Monitor Health and Status
Receiver
Monitor: Control Mode
Receiver
Transmitter
Monitor: Location and Navigation
Receiver
Transmitter
Monitor: Video Feed
Receiver
Monitor: Patient Secure Status
Receiver
Transmitter
MRA: Multirole Aircraft
Signal: Health and Status
Transmitter
Signal: Control Mode
Transmitter
Receiver
Signal: Location and Navigation
Transmitter
Receiver
Digital Service: Video Feed
Transmitter
ERU: Emergency Response Unit
Signal: Health and Status
Transmitter
Signal: Control Mode
Transmitter
Receiver
Signal: Location and Navigation
Transmitter
Receiver
Digital Service: Video Feed
Transmitter
Signal: Patient Secure Status
Transmitter
Receiver
MEA: Med-Evac Aircraft
Signal: Health and Status
Transmitter
Signal: Control Mode
Transmitter
Receiver
Signal: Location and Navigation
Transmitter
Receiver
Signal: Patient Secure Status
Transmitter
Receiver

Risk Analysis Key

#	Probability: Chance of Risk
1	Rare: Slim possibility of happening
2	Unlikely: Could occur but remote
3	Possible: Might happen
4	Likely: Expected to occur
5	Almost Certain: Expected regularly

#	Consequence: Impact on Mission
1	Insignificant: Not a cause for concern
2	Minor: Can cause delays or added costs
3	Moderate: Likely delays or costs
4	Major: Possible failure
5	Severe: Definite failure

Technical Risk Cube

	Technical Risks Matrix	1	2	3	4	5
		Consequence				
		Insignificant	Minor	Moderate	Major	Severe
5	Probability: Almost Certain					
4	Probability: Likely			T6		
3	Probability: Possible			T5		T1, T2, T3
2	Probability: Unlikely			T4		
1	Probability: Rare					

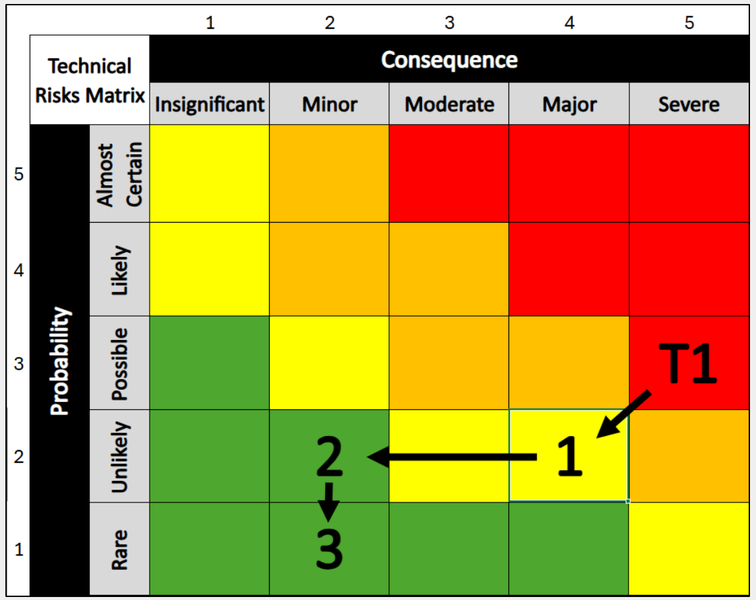
#	Risk Description
T1	Airborne Collision
T2	GCS Power Loss
T3	GCS Signal Loss
T4	Patient Unsecured
T5	FRS Radio Search Failure
T6	Unfavorable Package Orientation

Post-Mitigation Technical Risk Cube

Technical Risks Matrix		1	2	3	4	5
		Consequence				
		Insignificant	Minor	Moderate	Major	Severe
Probability 5	Almost Certain					
Probability 4	Likely					
Probability 3	Possible					
Probability 2	Unlikely		T6			
Probability 1	Rare		T1, T3, T4, T5	T2		

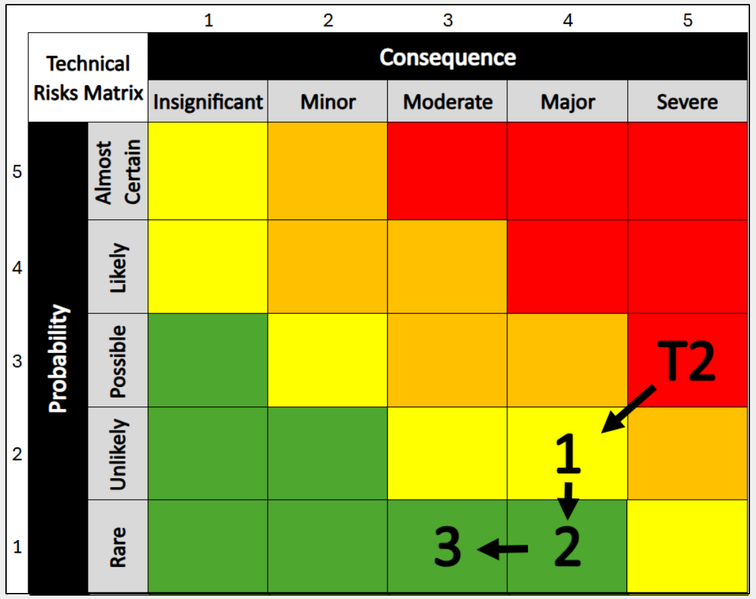
#	Risk Description
T1	Airborne Collision
T2	GCS Power Loss
T3	GCS Signal Loss
T4	Patient Unsecured
T5	FRS Radio Search Failure
T6	Unfavorable Package Orientation

Risk Mitigation – T1: Airborne Collision



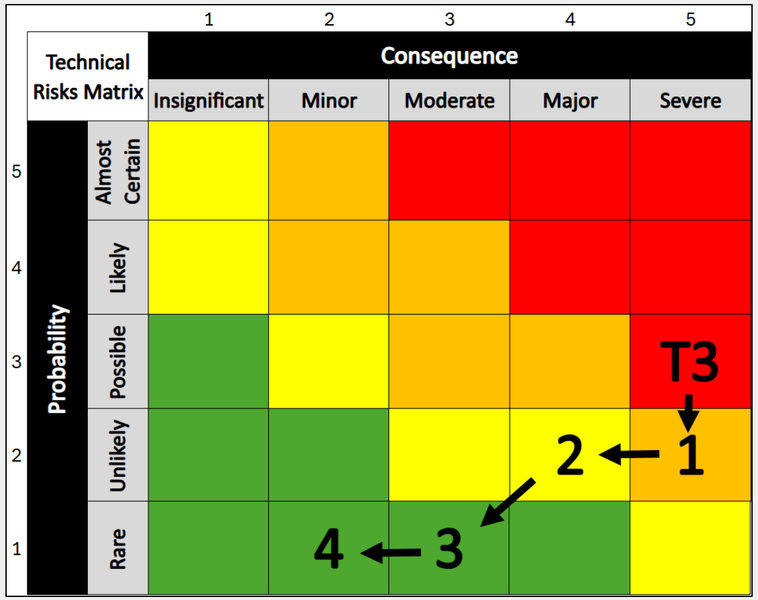
Risk Description	IF the flight paths of the MEA and MRA get too close to one another, THEN then an airborne collision may occur.		
ID	**Mitigation Plan**	**P**	**C**
1	Establish visual flight rules	2	5
2	Manually track the vehicles using binoculars	1	4
3	Develop software to track the distance between the MRA and MEA while airborne	1	2

Risk Mitigation – T2: GCS Power Loss



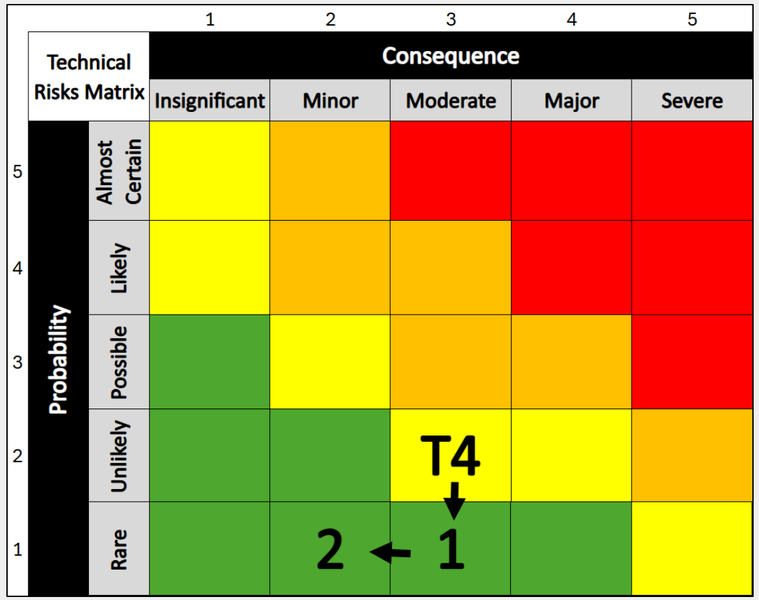
Risk Description	IF the GCS loses power, THEN the other actors are forced to operate independently.		
ID	**Mitigation Plan**	**P**	**C**
1	Have a dry run to ensure that all actors can operate independently through verbal communication	3	4
2	Bring a backup power source	1	4
3	Run the operation independently by verbal communication	1	3

Risk Mitigation – T3: GCS Signal Loss



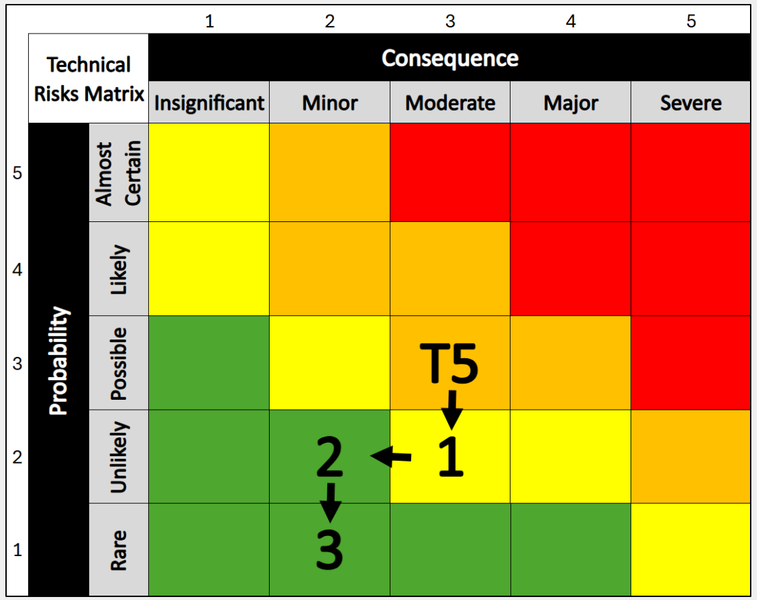
Risk Description	IF the GCS loses signal, THEN the GCS will initiate reconnection and command an emergency stop/loiter.		
ID	**Mitigation Plan**	**P**	**C**
1	Conduct range tests to identify areas with potential connection interference	2	5
2	Reattempt connection and continue to last known waypoint, then loiter	2	4
3	Change connection frequency	1	3
4	Notify operators of connection failure and operate independently (without GCS)	1	2

Risk Mitigation – T4: Patient Unsecure



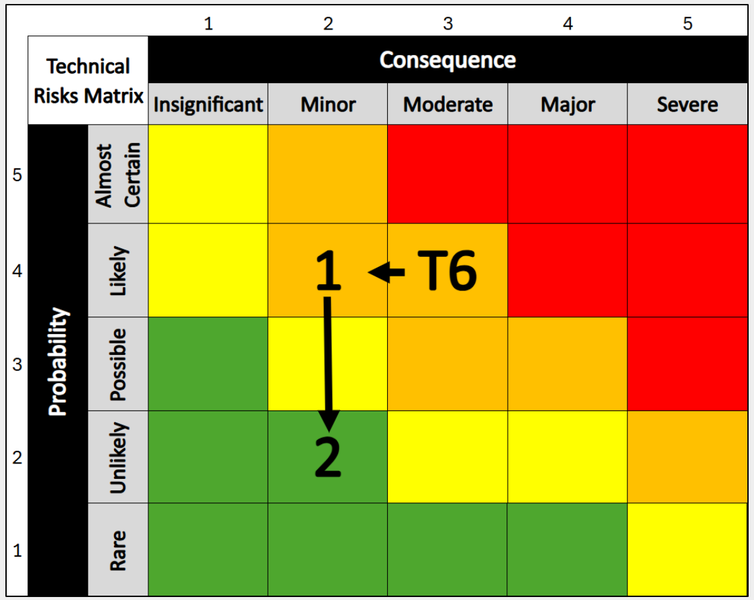
Risk Description	IF the ambulatory/immobile patient becomes unsecure during navigation, THEN recovering the patient will cause a delay		
ID	**Mitigation Plan**	**P**	**C**
1	Reduce vehicle speed when navigating weather conditions/uneven terrain	1	3
2	Switch to manual control and recover patient when loss of patient is detected	1	2

Risk Mitigation – T5: FRS Radio Search Failure



Risk Description	IF the RF search system fails to accurately locate the patient's FRS signal within the 10ft radius THEN the MRA may drop the medical supply package away from the intended target location.		
ID	**Mitigation Plan**	**P**	**C**
1	Conduct extensive tests to calibrate and optimize the radio frequency search system	2	3
2	Switch to manual mode, using the FPV camera to confirm drop location	1	2
3	Loiter drop area to aid ERU in locating and retrieving the medical supply package	1	2

Risk Mitigation – T6: Unfavorable Package Orientation



Risk Description	IF the Medical Supply Package is obstructed or lands in an unfavorable orientation, THEN the ERU will not be able to retrieve the package.		
ID	**Mitigation Plan**	**P**	**C**
1	Perform a dry run to test the retrieval mechanism in different orientations	4	2
2	Use the ERU's mechanism to change the orientation/position of the Medical Supply Package	2	2